MPCs, MPEs and Multipliers: A Trilemma for New Keynesian Models

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Three Macroeconomic Objects

- **MPC:** The amount by which consumption increases in response to a one-time transitory increase in income.
  - A micro moment for discriminating across models.
  - Used to calibrate macroeconomic models of consumption behaviour.
- **MPE:** The negative of the response of earned income to a one-time transitory increase in income. Labour market equivalent of MPC.

> MPC & MPE capture two adjustment-margins agents have when faced with an income shock: they can consume more or work less.

- **Fiscal Multiplier:** The response of GDP to a unit change in government spending. Depends on the response of monetary policy.

Three Empirical Facts

- Fact 1: The average MPC is about 0.25 quarterly and 0.5 annually.
- Fact 2: The annual MPE lies between 0 and 0.04
- Fact 3: Fiscal Multipliers lie between 0.6 and 2.0 when monetary policy is accommodative, for both impact and cumulative multipliers.

The HANK model

- Use a HANK model with frictionless labour supply (Aiyagari and McGrattan 1998).
  - Praised for matching MPCs (Fact 1).
- The household solves:

$$V_t(a,e) = max_{c,n,a'} U(c,n) + \beta E[V_{t+1}(a',e')|e]$$

$$s.t. \quad c + a' = (1-\tau_t^w)w_t e n + T_t + (1+r_t)a$$

$$a' \geq \underline{a}$$

- where e is a household idiosyncratic risk to productivity.

The MPC - MPE relationship

- HANK models usually apply **separable preferences** such that:

$$U(c,n) = u(c) - v(n)$$

- Alternative preference specifications (which dampen wealth effects on labour supply) are **GHH preferences** (Greenwood et al. 1988):

$$U(c,n) = u(c - v(n))$$

The Complementarity Index

- The paper defines a new **complementarity index (CI)**:

$$\mathrm{CI} \equiv \frac{\partial c(\lambda, \tilde{w})}{\partial \tilde{w}} / \tilde{w} \frac{\partial n(\lambda, \tilde{w})}{\partial \tilde{w}} = \frac{U_{nc}/U_n}{U_{cc}/U_c}$$

More detail on MPCs and MPEs

- The marginal propensity to consume out of a transfer:

$$MPC \equiv \frac{\delta c_t(a, e; T_t)}{\delta T_t}$$

- The marginal propensity to earn out of a transfer:

$$MPE \equiv -\tilde{w}\frac{\delta n_t(a, e; T_t)}{\delta T_t}$$

- Using these two definitions, we get:

$$\frac{MPE}{MPC} = \frac{\tilde{w}n}{c} * \frac{Frisch}{EIS} * (1 - CI)$$

$$\boxed{\Rightarrow MPE \approx MPC * (1 - CI)}$$

The Trilemma Strikes Back

$$MPE \approx MPC * (1 - CI)$$

- For separable preferences CI $= 0$.
  - Implies that for most agents, MPC and MPE are approximately the same. But that does not hold true in data (MPCs are high, MPEs are low).
- For GHH preferences CI $= 1$.
  - But that will lead to high fiscal multipliers, violating Fact 3.
  - MPE $= 0$

The CI Index & Fiscal Multipliers

- Rely on a simple Representative Agent Model for heuristic purposes.
  - RANK models do not fit the data on MPCs (Fact 1).
  - But clarify the link between the CI & Fiscal Multipliers.
- RANK is a special case of the baseline HANK model. Heterogeneity or idiosyncratic income risk are removed by setting:

$$e = 1 \text{ and } \underline{a} = -\infty.$$

- Consumption behaviour of the RA are characterized by Euler and Labour Supply equations, shown here under perfect foresight:

$$U_c\left(c_t, n_t\right) = \beta\left(1 + r_t^e\right) U_c\left(c_{t+1}, n_{t+1}\right)$$

$$\left(1 - \tau_t^w\right) w_t = -\frac{U_n\left(c_t, n_t\right)}{U_c\left(c_t, n_t\right)}$$

RANK

- On the supply side, the RANK model follows the textbook structure.
  - Final goods produced competitively as a CES aggregate of a continuum of intermediary goods with elasticity ϵ_p.
  - Intermediary goods produced by monopolists attempting to maximise the present value of profits.
  - Quadratic price adjustment costs, as in Rotemberg (1982).
- The symmetric equilibrium, where intermediate good producers set same prices & quantities, is characterized by an aggregate production function:

$$Y_t = f\left(N_t\right)$$

- New Keynesian Philipps Curve:

$$\log\left(1+\pi_t\right) = \kappa_p\left(\frac{w_t}{f'\left(N_t\right)} - \frac{\epsilon_p - 1}{\epsilon_p}\right) + \frac{1}{1+r_t^e}\frac{Y_{t+1}}{Y_t}\log\left(1+\pi_{t+1}\right)$$

- Labor wedge, as in Chari, Kehoe, and McGrattan (2007):

$$\tau_t \equiv 1 + \frac{U_n\left(C_t, N_t\right)}{U_c\left(C_t, N_t\right)}\frac{1}{f'\left(N_t\right)}$$

Fiscal Multiplier in RANK

- The Fiscal authority purchases G_t of the final good in every period, financed by changes in taxes or transfers.
- Market clearing requires that:

$$Y_t = C_t + G_t + \frac{\epsilon_p}{2\kappa_p} \log\left(1 + \pi_t\right)^2 Y_t$$

- PROPOSITION 2 – assuming a constant real rate:

$$\frac{dY_t}{dG_s} = \frac{1}{1 - (1 - \tau)\mathrm{CI}} \cdot \mathbf{1}_{s=t}$$

- Separable preferences $\Rightarrow \mathrm{CI} = 0 \Rightarrow \frac{dY_t}{dG_s} = 1$
- GHH preferences $\Rightarrow \mathrm{CI} = 1 \Rightarrow \frac{dY_t}{dG_s} = \frac{1}{\tau}$, with $\tau < 1$
- Steady-state labor wedge, as in Nakamura and Steinsson (2014):

$$\tau = 1 - \left(1 - \tau^w\right) \frac{\epsilon_p - 1}{\epsilon_p}$$

A preliminary outlook on the Trilemma

- Preliminary calculations reveal the tension in jointly accounting for Facts 1, 2 & 3.
- Matching 1 & 2, with sensible values for the Frisch & EIS, gives:

$$\text{CI} \geq 0.92$$

↪ *Halving the Frisch or doubling the EIS, this can be loosened to:*

$$\text{CI} \geq 0.84$$

- Matching 3 in a RANK model, with conservative choices for tax rates and goods elasticities, gives:

$$\text{CI} \leq 0.88$$

- This tension is exacerbated once we move to HANK models, which generate high MPCs and correspondingly higher fiscal multipliers. The next section verifies this hypothesis quantitatively.

The Trilemma for HANK Models

A quick roadmap:

- Introducing the HANK with "GHH-Plus" Preferences

  4 parts to the model:

  - HH
  - Asset Markets
  - Firms
  - Government

- Calibrating the HANK
- The MPE/ Multiplier Dilemma
- Parameters other than Complementarity
- Alternative Monetary and Fiscal Policies

The HANK and "GHH-Plus" Preferences

The Household

GHH-Plus Utility function:

$$U(c,n) = \frac{1}{1-\sigma}\left(c - \varphi\alpha\frac{n^{1+v}}{1+v}\right)^{1-\sigma} - \varphi(1-\alpha)\frac{n^{1+v}}{1+v}$$

GHH-Plus preferences are useful because:

- Nest both seperable and GHH preferences for $\alpha = 0$ and $\alpha = 1$ respectively.
- Deliver a "useful" expression for the CI:

$$\mathrm{CI}(e,a) = \frac{\alpha U_c(e,a)}{\alpha U_c(e,a) + 1 - \alpha} \in [0,1]$$

See how:

$$\alpha = 0 \Rightarrow CI = 0 \;\wedge\; \alpha = 1 \Rightarrow CI = 1$$

The HANK and "GHH-Plus" Preferences

The Household

GHH-Plus FYI:

- They are not the only type of preferences that nest both seperable and GHH!
- Bilbiie (2020) shows this by suggesting a different "quasi-seperable" utility function.
- More generally, Bilbiie (2020) is a good reference:

"If you are reading this, chances are you are a macroeconomist who often needs to choose a utility function—most likely, specifying household preferences over consumption and hours worked, $U(C, N)$. What utility function should a macroeconomist use?"

The HANK and "GHH-Plus" Preferences

Asset Markets

There are 2 assets in positive net supply:

1. Public debt
2. Equity in intermediate goods firms

Financial market equilibrium requires both assets to offer the same ex-ante return s.t.

$$1 + r_t^e = \mathbb{E}_t \left[\frac{d_{t+1} + p_{t+1}}{p_t} \right]$$

Assumption: all households hold the same portfolio of bonds and equity. Thus,

$$1 + r_t = \underbrace{\frac{p_{t-1}}{p_{t-1} + B_{t-1}}}_{\text{equity share}} \frac{d_t + p_t}{p_{t-1}} + \underbrace{\frac{B_{t-1}}{p_{t-1} + B_{t-1}}}_{\text{bond share}} (1 + r_{t-1}^e)$$

The HANK and "GHH-Plus" Preferences

Firms

On the supply side, the HANK model follows the textbook structure.

- Final goods are produced competitively as a CES aggregate of a continuum of intermediate goods with elasticity ϵ_p.
- Intermediate goods produced by monopolists attempting to maximise the present value of profits.
- Quadratic price adjustment costs, as in Rotemberg (1982).

The intermediate good production function:

$$f\left(N_t\right) = ZN_t - F$$

- is a linear function of aggregate labour scaled by stochastic productivity Z, minus fixed costs F

The HANK and "GHH-Plus" Preferences

Firms

Elasticity and equity play two distinct roles in the model:

- Elasticity affects the value of the labour wedge and therefore the fiscal multiplier

$$\tau = 1 - (1 - \tau^w)\,\frac{\epsilon_p - 1}{\epsilon_p} \qquad\qquad \frac{dY_t}{dG_s} = \frac{1}{1 - (1-\tau)\mathrm{CI}} \cdot \mathbf{1_{s=t}}$$

- Firm equity is an asset held by HH.
  - Determines overall degree of liquidity in economy.
  - Key determinant of MPCs.

Aggregate firm dividends:

$$d_t = Y_t - w_t N_t - \frac{\epsilon_p}{2\kappa}\log\left(1+\pi_t\right)^2 Y_t$$

The HANK and "GHH-Plus" Preferences

The Role of Government

- Government budget constraint:

$$B_t + \tau_t^w w_t N_t = \left(1 + r_{t-1}^e\right) B_{t-1} + G_t + T$$

- Fiscal policy rule, following Auclert and Rognlie (2018):

$$B_t - B_{ss} = \rho_B \left(B_{t-1} - B_{ss} + G_t - G_{ss}\right)$$

s.t. $\rho_B \in [0, 1)$ is the "persistence of debt"

- Monetary policy rule:

$$r_t^e \equiv r_{ss}$$

- Finally, goods market clearing requires:

$$Y_t = C_t + G_t + \frac{\epsilon_p}{2\kappa_p} \log\left(1 + \pi_t\right)^2 Y_t$$

Calibration

Remember:

- Authors are using a HANK model with canonical supply side, sticky prices, and flexible wages.
- Goal is to study the tradeoff between Facts 2 & 3, conditional on Fact 1.
- Calibration follows from McKay, Nakamura, and Steinsson (2016) Song et al. (2019) and Auclert and Rognlie (2018)

Importantly,

- HANK models with a plausible supply of total assets cannot generate such large MPCs without additional features, such as illiquid assets or discount factor heterogeneity.
- Assets are liquid in this model in this model, hence the authors look to create discount factor heterogeneity, $\beta_2 < \beta_1$.

Calibration

parameter	name	value/target
$1/\nu$	Frisch elasticity	0.5
ρ_e	persistence of income process	0.966
τ^w	wage tax rate	0.334
T	transfer	$0.143 \cdot wN$
ϵ_p	elasticity of substitution	7
κ_p	NKPC slope	0.01
B	government bonds	$0.55 \cdot 4Y$
ρ_B	persistence of public debt	0.9
$\underline{a}$	borrowing constraint	0
β_1	upper discount factor	$r = 0.02/4$
β_2	lower discount factor	MPC $= 0.25$
$1/\sigma$	U_c curvature	average EIS $= 0.5$
φ	disutility of labor	$N = 1$
Z	aggregate labor productivity	$Y = 1$
σ_e	std of income shocks	$\text{Var}\left[\log(n_i e_i)\right] = 0.92^2$
F	fixed cost	$p = 0.85 \cdot 4Y$

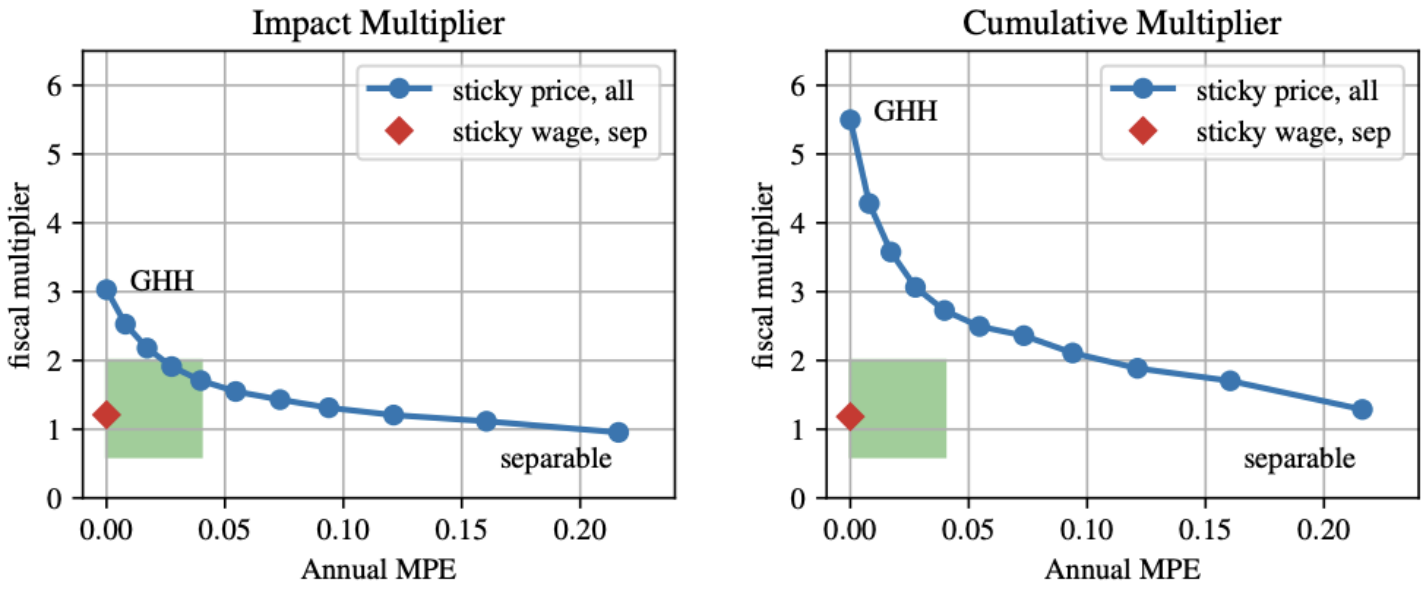
The upper panel lists directly calibrated parameters. The lower panel lists parameters calibrated to match some target, whose values will depend on the choice of α. Separable preferences: $(\beta_1, \beta_2) = (0.978, 0.923)$. GHH: $(\beta_1, \beta_2) = (0.987, 0.970)$.

MPE/ Multiplier Dilemma

- Idiosyncratic labor productivity e and gov spending G follow AR(1) processes with 0.966 and 0.9 respective persistences.
  - AR(1): current value of the variable is a linear function of its previous value and a random error term.

$$X_t = \phi X_{t-1} + \varepsilon_t$$

- It takes high complementarity to match Fact 2
- It takes low complementarity to match Fact 3



Parameters other than Complementarity

What about a higher Elasticity of Intertemporal Substitution (1.0)?

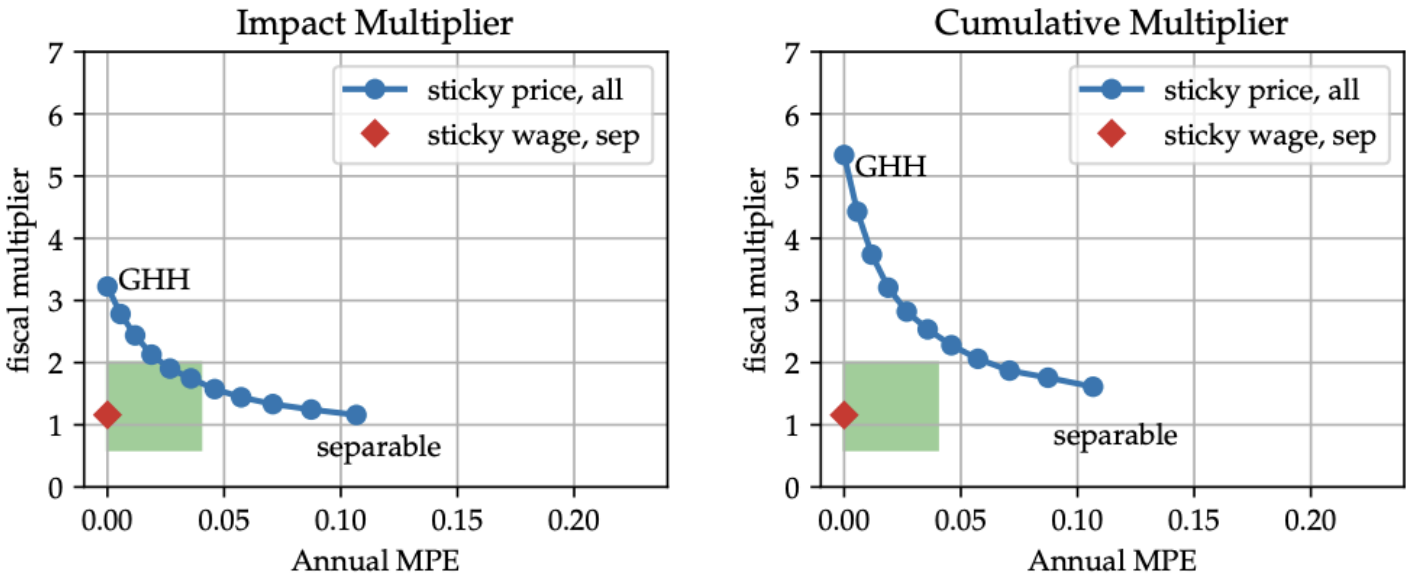


Figure: Higher Elasticity of Intertemporal Substitution (1.0)

Parameters other than Complementarity

What about a Lower Frisch Elasticity (0.25)?

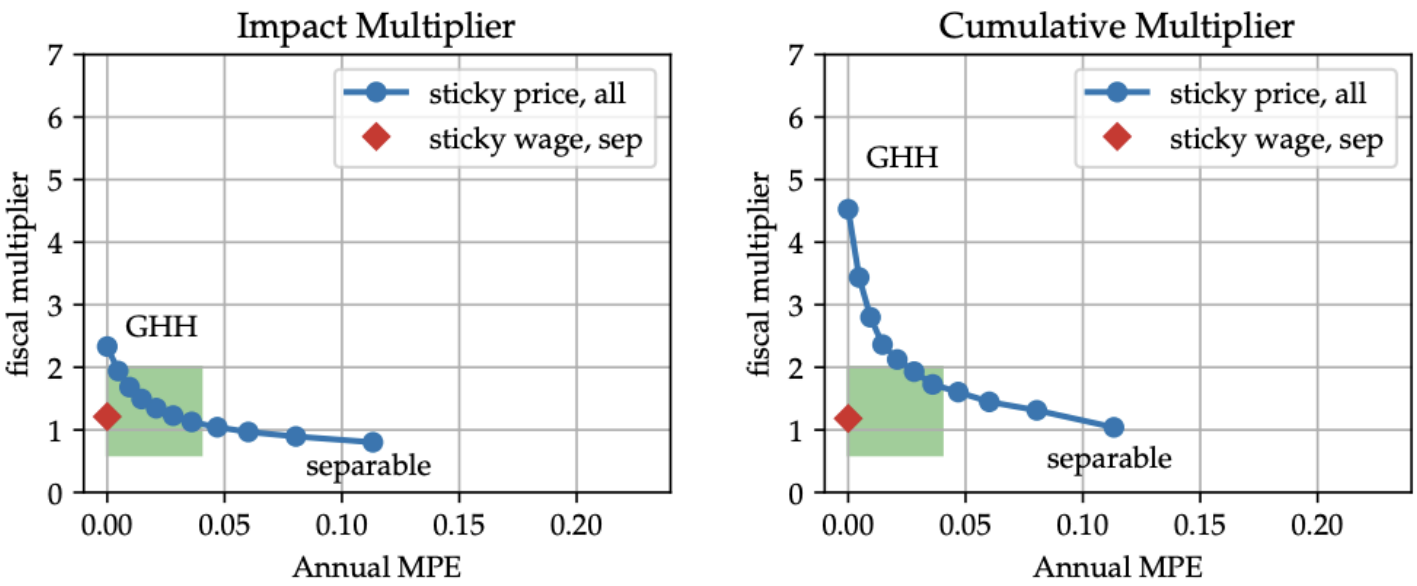


Figure: Lower Frisch Elasticity (0.25)

Parameters other than Complementarity

What about a higher markup (50%)?

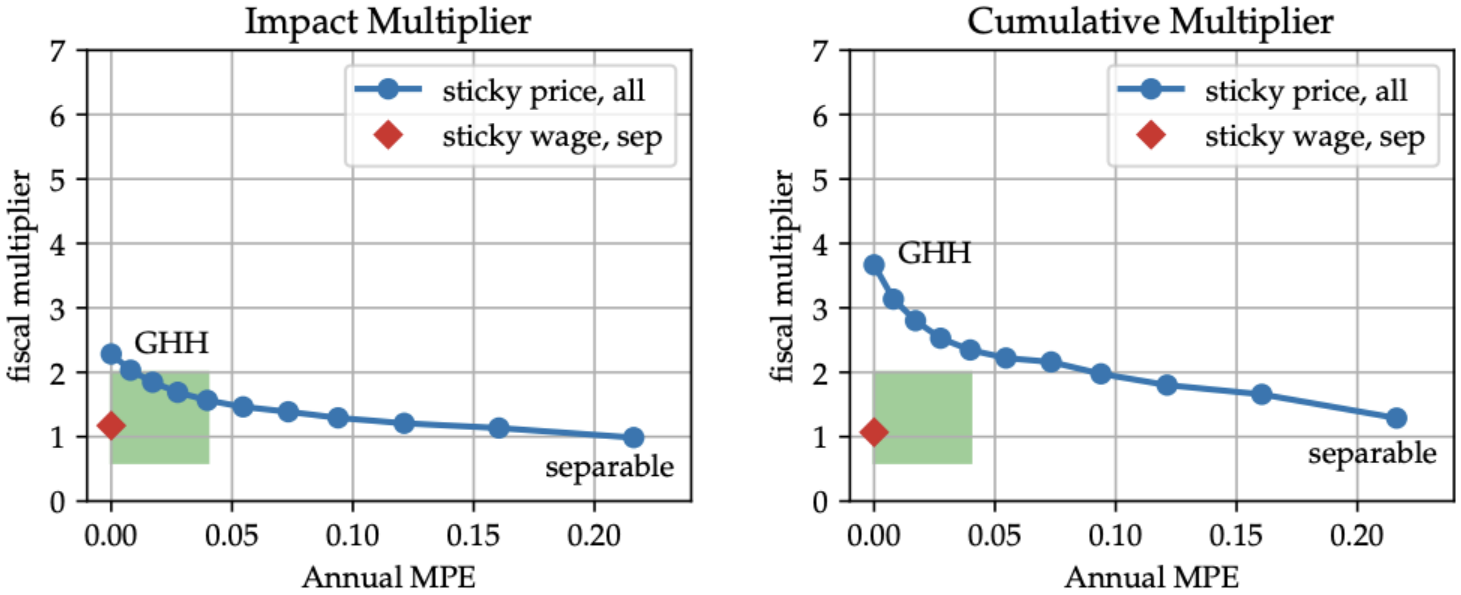


Figure: Higher Markup (50%)

Parameters other than Complementarity

What about a progressive income tax?

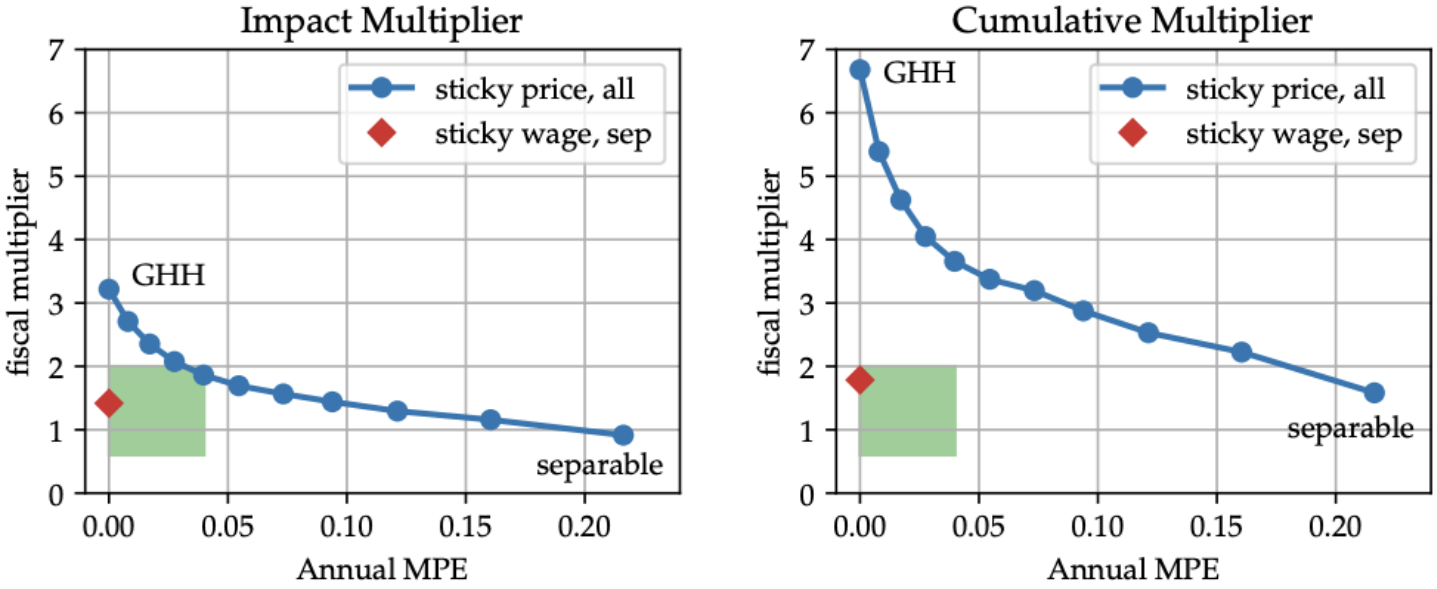


Figure: Progressive Tax

Alternative Fiscal and Monetary Policies

Playing with Monetary Policy:

- Taylor Rule: $i_t = r + \phi \pi_t$ s.t. the coefficient on inflation $\phi = 1.25$
- An always accomodative policy (constant peg, constant-r)
- 3 year peg, followed by Taylor rule

Playing with Fiscal Policy:

- $\rho = 0$
- $\rho = 0.9$ (baseline)
- $\rho = 0.95$

Alternative Fiscal and Monetary Policies

Interestingly:

- The timing of debt financing is irrelevant for the multiplier.
- All sticky wage multipliers match Fact 3!

	Monetary			Fiscal		
	Taylor rule	constant-r	3-year peg	$\rho_B = 0$	$\rho_B = 0.9$	$\rho_B = 0.95$
Separable	0.77	1.29	73.92	1.29	1.29	1.29
$\alpha = 0.5$	1.14	2.49	3.49	2.49	2.49	2.49
GHH	2.39	5.50	6.00	5.50	5.50	5.50
Sticky Wage	0.93	1.18	1.46	0.94	1.18	1.59

Figure: Cumulative Fiscal Multipliers with Alternative Monetary and Fiscal Policies

Resolving the Trilemma

- Demand-determined labor, nominal wage-stickiness, flexible prices, as in Erceg, Henderson, and Levin (2000).
- Households supply a continuum of differentiated labor services $\mathrm{L}_t(k), k \in [0,1]$, which are imperfect substitutes with elasticity ϵ_w.
- Combined by firms into a final labor input for production:

$$L_t = \left[\int_0^1 \mathrm{L}_t(k)^{\frac{1}{1+\theta_w}} dh \right]^{1+\theta_w}$$

- Wages w_{kt} are set by unions specific to each labor variety k. Firms take wages as given. Cost minimization, given wages and total labor N, yields demand schedules for each k:

$$N_{kt} = \left(\frac{w_{kt}}{w_t} \right)^{-\epsilon_w} N_t$$

- Assumption: firms set quantities for each labor variety, workers comply. If workers could adjust their labor supply, they would choose it such as to equate the real wage to the MRS.

Resolving the Trilemma

- With workers taken off their labor supply curves, by construction:

$$\text{MPE} = 0$$

$\hookrightarrow$ Workers are effectively deprived of one margin of adjustment to one-time income shocks.

- Previously, we needed a high value for CI to drive down the MPE relative to the MPC.
- Now, CI may be set to a low value, generating multipliers in line with Fact 3. Remember:

$$\frac{dY_t}{dG_s} = \frac{1}{1-(1-\tau)\text{CI}} \cdot \mathbf{1}_{s=t}$$

- Prices are assumed flexible, such that intermediate good producers charge their preferred markup, implying a constant real wage.

Resolving the Trilemma

- Separable preferences $\Rightarrow \alpha = 0$
- Calibration chosen to meet Fact 1 by construction.
- With demand-determined labor Fact 2 also met by construction.

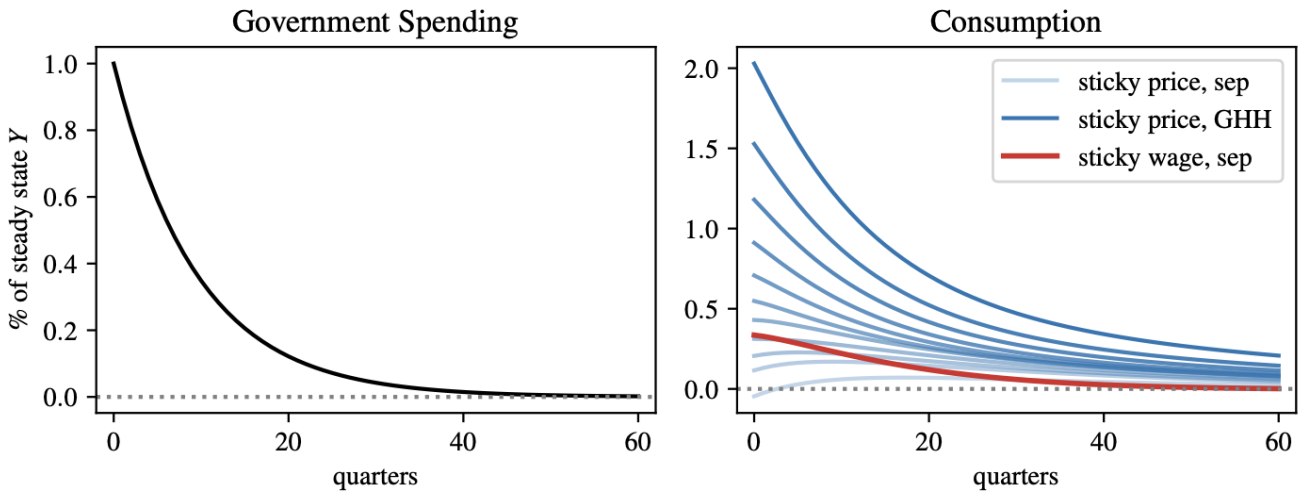


Figure: Government Spending and the Response of Private Consumption

Sticky wages, not Sticky Prices?

- Recent macro-econometric models have typically followed Erceg, Henderson, and Levin (2000), and assumed that both prices and nominal wages are slow to adjust.
- Christiano, Eichenbaum, and Evans (2005) find that nominal wage rigidity is more important than nominal price rigidity for explaining the dynamic effects of monetary policy shocks.
- Sticky wages resolve the issue of countercyclical price mark-ups: in NK models, a demand shock raises output and marginal cost, but sinces prices are sticky, price markups over marginal costs fall Nekarda and Ramey (2020).

Critiques

- An abstract exercise
  - Do little to provide microfoundations for any of their calibrations.

↪ However, doing so allows them to effectively isolate the tension between facts 1, 2 and 3.

- Do not motivate their sticky-wage solution
  - Do not delve into microfoundations for abandoning sticky prices and opting for sticky wages instead.

↪ However, admit that more explicit microfoundations are needed and insist that their main contribution is to demonstrate the dilemma.

References I

- Auclert, Adrien and Matthew Rognlie (2018). *Inequality and aggregate demand*. Tech. rep. National Bureau of Economic Research.
- Bilbiie, Florin O (2020). "A GHH-CRRA Utility for Macro: Complementarity". In: *Income, and Substitution*.
- Chari, Varadarajan V, Patrick J Kehoe, and Ellen R McGrattan (2007). "Business cycle accounting". In: *Econometrica* 75.3, pp. 781–836.
- Christiano, Lawrence J, Martin Eichenbaum, and Charles L Evans (2005). "Nominal rigidities and the dynamic effects of a shock to monetary policy". In: *Journal of political Economy* 113.1, pp. 1–45.

References II

- Erceg, Christopher J, Dale W Henderson, and Andrew T Levin (2000). "Optimal monetary policy with staggered wage and price contracts". In: *Journal of monetary Economics* 46.2, pp. 281–313.
- McKay, Alisdair, Emi Nakamura, and Jón Steinsson (2016). "The power of forward guidance revisited". In: *American Economic Review* 106.10, pp. 3133–3158.
- Nakamura, Emi and Jón Steinsson (2014). "Fiscal stimulus in a monetary union: Evidence from US regions". In: *American Economic Review* 104.3, pp. 753–792.
- Nekarda, Christopher J and Valerie A Ramey (2020). "The cyclical behavior of the price-cost markup". In: *Journal of Money, Credit and Banking* 52.S2, pp. 319–353.
- Rotemberg, Julio J (1982). "Sticky prices in the United States". In: *Journal of political economy* 90.6, pp. 1187–1211.

References III

Song, Jae et al. (2019). "Firming up inequality". In: *The Quarterly journal of economics* 134.1, pp. 1–50.